

姓名：

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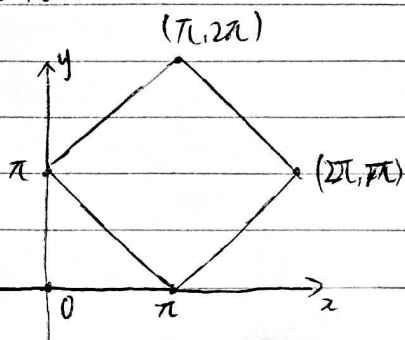
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練習題 16.6

1. (1)



$$D = \{(x, y) \mid |x - \pi| + |y - \pi| \leq \pi\}$$

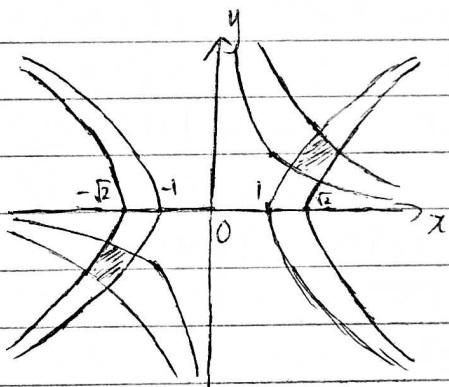
$$\text{設 } u = x - \pi, \quad v = y - \pi, \quad \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = |I| = 1$$

$$\iint_D (x - y)^2 \sin(x + y) \, dx \, dy = \iint_{|u| + |v| \leq \pi} (u - v)^2 \sin(u + v) \, du \, dv$$

$$= \iint_{|u| + |v| \leq \pi} (-u + v)^2 \sin(-u - v) \, du \, dv = - \iint_{|u| + |v| \leq \pi} (u - v)^2 \sin(u + v) \, du \, dv$$

$$\Rightarrow \iint_D (x - y)^2 \sin(x + y) \, dx \, dy = 0$$

(2)



$$\text{令 } u = x^2 - y^2, \quad v = xy$$

$$(u, v) \in [1, 2]^2$$

$$u = x^2 - \frac{v^2}{x^2} \Leftrightarrow x^4 - ux^2 - v^2 = 0$$

$$x^2 = \frac{u + \sqrt{u^2 + 4v^2}}{2}, \quad y^2 = \frac{\sqrt{u^2 + 4v^2} - u}{2}$$

$$\text{故 } D_1 = \{(x, y) \mid x = \sqrt{\frac{\sqrt{u^2 + 4v^2} + u}{2}}, y = \sqrt{\frac{\sqrt{u^2 + 4v^2} - u}{2}}\}$$

$$\frac{\partial(x, y)}{\partial(u, v)} \Big|_{D_1} = \begin{vmatrix} \frac{u}{\sqrt{u^2 + 4v^2}} + 1 & \frac{4v}{\sqrt{u^2 + 4v^2}} \\ 2\sqrt{2} \sqrt{\sqrt{u^2 + 4v^2} + u} & 2\sqrt{2} \sqrt{\sqrt{u^2 + 4v^2} + u} \\ \frac{u}{\sqrt{u^2 + 4v^2}} - 1 & \frac{4v}{\sqrt{u^2 + 4v^2}} \\ 2\sqrt{2} \sqrt{\sqrt{u^2 + 4v^2} - u} & 2\sqrt{2} \sqrt{\sqrt{u^2 + 4v^2} - u} \end{vmatrix}$$

$$D_2 = \{(x, y) \mid x = -\sqrt{\frac{\sqrt{u^2 + 4v^2} + u}{2}}, y = -\sqrt{\frac{\sqrt{u^2 + 4v^2} - u}{2}}\}$$

$$= \frac{4v}{8\sqrt{u^2 + 4v^2} - u^2} \times 2 = \frac{1}{2\sqrt{u^2 + 4v^2}} = \frac{\partial(x, y)}{\partial(u, v)} \Big|_{D_2}$$

所以 $\iint_D (x^2+y^2) dx dy = \iint_{[1,2]^2} \sqrt{u^2+4v^2} \cdot \frac{1}{2\sqrt{u^2+4v^2}} du dv = \frac{1}{2} \iint_{[1,2]^2} du dv = \frac{1}{2}$

2. (1) 記 $a_1x+b_1y+c_1=r\cos\theta$, $a_2x+b_2y+c_2=r\sin\theta$, $r \in [0,1]$, $\theta \in [0,2\pi)$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}^{-1} \begin{pmatrix} r\cos\theta - c_1 \\ r\sin\theta - c_2 \end{pmatrix} = \frac{1}{a_1b_2-a_2b_1} \begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix} \begin{pmatrix} r\cos\theta - c_1 \\ r\sin\theta - c_2 \end{pmatrix}$$

$$\begin{pmatrix} \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial r} \end{pmatrix} = \frac{1}{a_1b_2-a_2b_1} \begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \quad \begin{pmatrix} \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \theta} \end{pmatrix} = \frac{r}{a_1b_2-a_2b_1} \begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix} \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

$$\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}^{-1} \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

$$\left| \frac{\partial(x,y)}{\partial(r,\theta)} \right| = \left| \frac{r}{a_1b_2-a_2b_1} \right| = \frac{r}{|a_1b_2-a_2b_1|}$$

$$\iint_D dx dy = \int_0^{2\pi} d\theta \int_0^1 \frac{r^2}{|a_1b_2-a_2b_1|} dr = 2\pi \cdot \frac{1}{3} \cdot \frac{1}{|a_1b_2-a_2b_1|} = \frac{2\pi}{3|a_1b_2-a_2b_1|}$$

(2) 記 $x=u^2$, $y=v^2$, $D = \{(x,y) \mid \sqrt{x}+\sqrt{y} \leq \sqrt{a}\}$, $\tilde{D} = \{(u,v) \mid u+v \leq \sqrt{a} \text{ 且 } u,v \geq 0\}$

$$\iint_D dx dy = \iint_{\tilde{D}} |4uv| du dv = 4 \iint_{\tilde{D}} uv du dv = 4 \int_0^{\sqrt{a}} du \int_0^{\sqrt{a}-u} dv$$

$$= 4 \int_0^{\sqrt{a}} (\sqrt{a}-u) du = 4 \left(a - \frac{1}{2} u^2 \Big|_0^{\sqrt{a}} \right) = 2a$$

3. 証: 記 $D = \{(x,y) \mid |x|+|y| \leq 1\}$, $u=x+y$, $v=x$, $\left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \left| \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} \right| = 1$

$$\iint_D f(x+y) dx dy = \int_{-1}^1 du \int_{\frac{u-1}{2}}^{\frac{u+1}{2}} f(u) dx = \left(\int_{-1}^1 f(u) du \right) \cdot \left(\int_{\frac{u-1}{2}}^{\frac{u+1}{2}} dx \right) = \int_{-1}^1 f(u) du$$

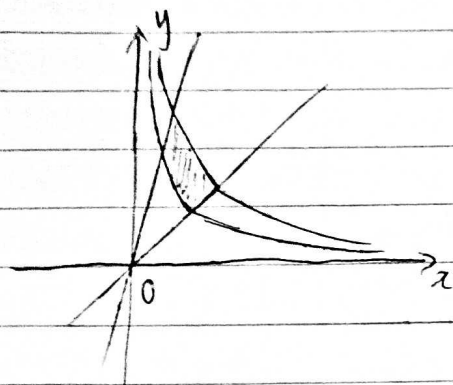
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設 $xy = u$, $y = vx$, $(u, v) \in [1, 2] \times [1, 4]$

$$x = \sqrt{\frac{u}{v}}, \quad y = \sqrt{uv}$$

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \left| \begin{vmatrix} \frac{1}{2\sqrt{v}} & -\frac{1}{2}\sqrt{\frac{u}{v^3}} \\ \frac{\sqrt{v}}{2\sqrt{u}} & \frac{\sqrt{u}}{2\sqrt{v}} \end{vmatrix} \right| = \frac{1}{4} \cdot \frac{1}{v} + \frac{1}{4} \cdot \frac{1}{v} = \frac{1}{2v}$$

$$\iint_D f(xy) dx dy = \iint_{[1, 2] \times [1, 4]} f(u) \cdot \frac{1}{2v} du dv$$

$$= \frac{1}{2} \left(\int_1^2 f(u) du \right) \left(\int_1^4 \frac{1}{v} dv \right) = \ln 2 \cdot \left(\int_1^2 f(t) dt \right)$$

(2) 設 $x = r \cos \theta$, $y = r \sin \theta$, $r \geq 0$

$$x^2 + y^2 \leq Rx \Leftrightarrow r^2 \leq Rr \cos \theta \Leftrightarrow r \leq R \cos \theta$$

$$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad r \in [0, R \cos \theta]$$

$$\left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| = r$$

$$\begin{aligned} \iint_{x^2+y^2 \leq Rx} \sqrt{R^2 - x^2 - y^2} dx dy &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{R \cos \theta} r \sqrt{R^2 - r^2} dr = - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{3} (R^2 - r^2)^{\frac{3}{2}} \Big|_0^{R \cos \theta} d\theta \\ &= -\frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (R^3 \sin^3 \theta - R^3) d\theta = \frac{\pi}{3} R^3 \end{aligned}$$

(4) 設 $x = r \cos \theta$, $y = r \sin \theta$, $r \geq 0$

$$x^2 + y^2 \leq x + y \Leftrightarrow r \leq \cos \theta + \sin \theta = \sqrt{2} \cos\left(\theta - \frac{\pi}{4}\right), \quad \theta \in \left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

$$\begin{aligned} \iint_{x^2+y^2 \leq x+y} \sqrt{x^2 + y^2} dx dy &= \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} d\theta \int_0^{\sqrt{2} \cos(\theta - \frac{\pi}{4})} r^2 dr = \frac{1}{3} \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} 2\sqrt{2} \cos^3\left(\theta - \frac{\pi}{4}\right) d\theta = \frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\sqrt{2} \cos^3 \theta d\theta \\ &= \frac{2\sqrt{2}}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - \sin^2 \theta) d\sin \theta = \frac{2\sqrt{2}}{3} \left(2 - \frac{1}{3} \times 2\right) = \frac{8\sqrt{2}}{9} \end{aligned}$$

7. 設 $x = ar \cos \theta$, $y = br \sin \theta$, $\theta \in [0, \frac{\pi}{2}]$, $r \in [0, 1]$

$$\left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| = abr \Rightarrow \iint_D \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}} dx dy = \int_0^{\frac{\pi}{2}} d\theta \int_0^1 r \cdot ab dr = ab \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi ab}{4}$$

8. 設 $x = r \cos \theta$, $y = r \sin \theta$, $r \in [0, +\infty)$, $\theta \in [0, 2\pi)$

$$\text{記 } \iint_{[R, R]^2} e^{2xy - 2x^2 - y^2} dx dy = S(R), \quad \iint_{x^2 + y^2 \leq R^2} e^{2xy - 2x^2 - y^2} dx dy = V(R)$$

$$V(R) = \int_0^{2\pi} d\theta \int_0^R e^{2r^2 \cos \theta \sin \theta - 2r^2 \cos^2 \theta - r^2 \sin^2 \theta} r dr$$

$$= \frac{1}{2} \int_0^{2\pi} \frac{1}{\sin 2\theta - \cos 2\theta - 1} \cdot \left(e^{\frac{R^2}{2}(\sin 2\theta - \cos 2\theta - 1)} - 1 \right) d\theta$$

$$= \int_0^{2\pi} \frac{1}{2\sin 2\theta - \cos 2\theta - 3} \cdot \left(e^{\frac{R^2}{2}(2\sin 2\theta - \cos 2\theta - 3)} - 1 \right) d\theta$$

$$= \int_0^{2\pi} \frac{1}{3 - \sqrt{5} \sin(2\theta - \beta)} \left(1 - e^{\frac{R^2}{2}(1 - \sqrt{5} \sin(2\theta - \beta) - 3)} \right) d\theta \quad \left(\text{記 } \beta = \arctan \frac{1}{2} \right)$$

$$= \int_0^{2\pi} \frac{1}{3 - \sqrt{5} \sin(\theta - \beta)} \left(1 - e^{\frac{R^2}{2}(\sqrt{5} \sin(\theta - \beta) - 3)} \right) d\theta$$

$$= 2 \int_0^{2\pi} \frac{1}{3 - \sqrt{5} \sin \theta} \left(1 - e^{\frac{R^2}{2}(\sqrt{5} \sin \theta - 3)} \right) d\theta \quad \left(\text{設 } y = \tan \frac{\theta}{2}, dy = \frac{1}{2}(y^2 + 1) d\theta \right)$$

$$\text{計算 } \int_0^{2\pi} \frac{1}{3 - \sqrt{5} \sin \theta} d\theta = \int_0^{2\pi} \frac{1}{3 - \frac{2\sqrt{5} \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}} d\theta = \int_{-\infty}^{+\infty} \frac{1+y^2}{-\infty + 3y^2 - 2\sqrt{5}y} \cdot \frac{2dy}{y^2+1} = \int_{-\infty}^{+\infty} \frac{2}{3y^2 - 2\sqrt{5}y + 3} dy$$

$$= 6 \int_{-\infty}^{+\infty} \frac{1}{(3y - \sqrt{5})^2 + 4} dy = 2 \int_{-\infty}^{+\infty} \frac{1}{y^2 + 4} dy = \int_{-\infty}^{+\infty} \frac{1}{\left(\frac{y}{2}\right)^2 + 1} d\left(\frac{y}{2}\right) = \arctan \frac{y}{2} \Big|_{-\infty}^{+\infty} = \pi$$

$$\int_0^{2\pi} \frac{1}{3 - \sqrt{5} \sin \theta} \cdot e^{-\frac{R^2}{2}(3 - \sqrt{5} \sin \theta)} d\theta = \frac{2\pi}{(3 - \sqrt{5} \sin \xi) e^{\frac{R^2}{2}(3 - \sqrt{5} \sin \xi)}} \quad (\xi \in (0, 2\pi), \text{積分中值定理})$$

$$\text{所以 } V(R) = 2\pi - \frac{4\pi}{(3 - \sqrt{5} \sin \xi) e^{\frac{R^2}{2}(3 - \sqrt{5} \sin \xi)}}$$

$$\text{又知 } V(R) \leq S(R) \leq V(\sqrt{2}R), \text{ 令 } R \rightarrow +\infty, \text{ 則 } \left| \frac{4\pi}{(3 - \sqrt{5} \sin \xi) e^{\frac{R^2}{2}(3 - \sqrt{5} \sin \xi)}} \right| \leq \frac{4\pi}{(3 - \sqrt{5}) \cdot e^{\frac{R^2}{2}(1 - \sqrt{2})}} \rightarrow 0$$

$$\text{則 } \iint_{\mathbb{R}^2} e^{2xy - 2x^2 - y^2} dx dy = 2\pi$$

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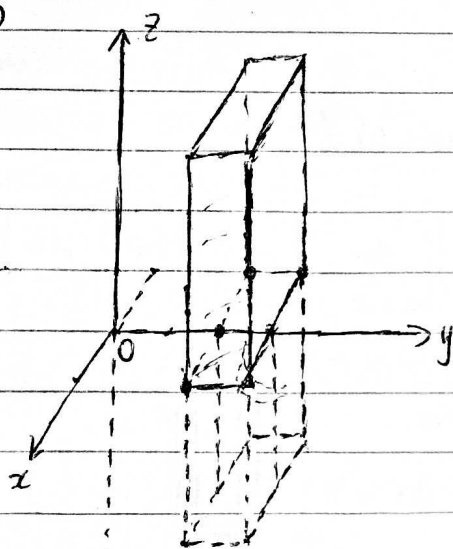
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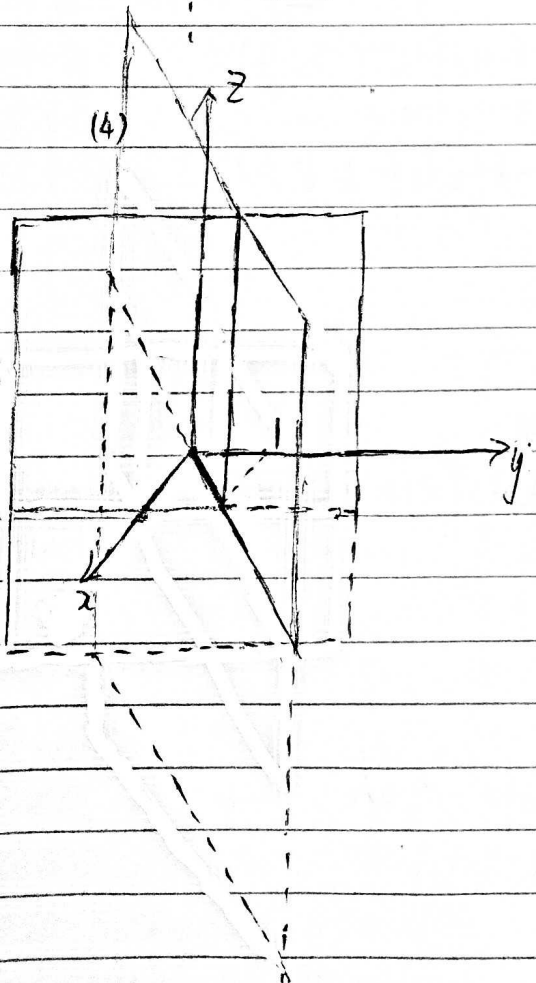
練習題 16.7

1. (3)



$$\begin{aligned} \int z dx dy dz &= \int_{-1}^0 dx \int_2^3 dy \int_{xy}^0 z dz + \int_0^1 dx \int_2^3 dy \int_0^{xy} z dz \\ &= \int_{-1}^0 dx \int_2^3 -\frac{1}{2}(xy)^2 dy + \int_0^1 dx \int_2^3 \frac{1}{2}(xy)^2 dy \\ &= -\frac{1}{2} \int_{-1}^0 x^2 dx \int_2^3 y^2 dy + \frac{1}{2} \int_0^1 x^2 dx \int_2^3 y^2 dy \\ &= \frac{1}{2} \left(\frac{1}{3} - 0 - \left(0 - \frac{1}{3}(-1)^3 \right) \right) \int_2^3 y^2 dy \\ &= 0 \end{aligned}$$

(4)



$$\begin{aligned} \int_V xy^2 z^3 dx dy dz &= \int_0^1 dx \int_0^2 dy \int_0^{xy} xy^2 z^3 dz \\ &= \int_0^1 x dx \int_0^2 y^2 \cdot \frac{1}{4}(xy)^4 dy = \frac{1}{4} \int_0^1 x^6 dx \int_0^2 y^6 dy \\ &= \frac{1}{28} \int_0^1 x^6 dx = \frac{1}{13 \times 28} = \frac{1}{364} \end{aligned}$$

$$2. (1) V = \{(x, y, z) \mid \frac{x^2}{2z} + \frac{y^2}{4z} \leq 1 \leq \frac{x^2}{z^2} + \frac{y^2}{4z^2}\}$$

$$2z \geq z^2 \text{ 或 } 4z \leq 4z^2 \Leftrightarrow 0 \leq z \leq 2$$

$$x^2 + \frac{y^2}{2} \leq 2z \text{ 且 } x^2 + \frac{y^2}{4} \geq z^2$$

$$4z^2 - 4x^2 \leq y^2 \leq 4z - 2x^2, \quad 1(z^2 - z) \leq x^2 \leq 2z$$

$$\text{當 } 0 \leq z \leq 1 \text{ 時, } 0 \leq x^2 \leq 2z \Leftrightarrow 0 \leq |x| \leq \sqrt{2z}$$

$$\text{當 } 1 \leq z \leq 2 \text{ 時, } x(z^2 - z) \leq x^2 \leq 2z \Leftrightarrow \sqrt{2z^2 - 2z} \leq |x| \leq \sqrt{2z}$$

$$\text{當 } 0 \leq |x| \leq z \text{ 時, } 2\sqrt{z^2 - x^2} \leq |y| \leq \sqrt{4z - 2x^2}$$

$$\text{當 } z \leq |x| \leq \sqrt{2z} \text{ 時, } 0 \leq |y| \leq \sqrt{4z - 2x^2}$$

$$\text{由對稱性知, } \int dx dy dz = 4 \int_0^1 dz \left(\int_0^z dx \int_{\sqrt{2z^2 - x^2}}^{\sqrt{4z - 2x^2}} dy + \int_z^{\sqrt{2z}} dx \int_0^{\sqrt{4z - 2x^2}} dy \right) + 4 \int_1^2 dz \left(\int_{\sqrt{2z^2 - 2z}}^z dx \int_{\sqrt{2z^2 - x^2}}^{\sqrt{4z - 2x^2}} dy + \int_z^{\sqrt{2z}} dx \int_0^{\sqrt{4z - 2x^2}} dy \right)$$

$$= 4 \int_0^1 dz \left(\int_0^z (\sqrt{4z - 2x^2} - \sqrt{2z^2 - x^2}) dx + \int_z^{\sqrt{2z}} \sqrt{4z - 2x^2} dx \right) + 4 \int_1^2 dz \left(\int_{\sqrt{2z^2 - 2z}}^z (\sqrt{4z - 2x^2} - \sqrt{2z^2 - x^2}) dx + \int_z^{\sqrt{2z}} \sqrt{4z - 2x^2} dx \right)$$

$$= 4 \int_0^1 dz \int_0^{\sqrt{2z}} \sqrt{4z - 2x^2} dx - 8 \int_0^1 dz \int_0^z \sqrt{2z^2 - x^2} dx + 4 \int_1^2 dz \int_{\sqrt{2z^2 - 2z}}^z \sqrt{4z - 2x^2} dx - 8 \int_1^2 dz \int_z^{\sqrt{2z}} \sqrt{2z^2 - x^2} dx$$

$$= 8 \int_0^1 \sqrt{2z} dz \int_0^{\sqrt{\frac{x}{2z}}} \sqrt{1 - \left(\frac{x}{\sqrt{2z}}\right)^2} d\left(\frac{x}{\sqrt{2z}}\right) - 8 \int_0^1 \frac{1}{2} dz \int_0^z \sqrt{1 - \left(\frac{x}{z}\right)^2} d\left(\frac{x}{z}\right) + 8 \int_1^2 z dz \int_{\sqrt{\frac{x}{2z}}}^{\sqrt{\frac{x}{2z}}} \sqrt{1 - \left(\frac{x}{\sqrt{2z}}\right)^2} d\left(\frac{x}{\sqrt{2z}}\right)$$

$$- 8 \int_1^2 z dz \int_{\sqrt{\frac{x}{2z}}}^z \sqrt{1 - \left(\frac{x}{z}\right)^2} d\left(\frac{x}{z}\right)$$

$$= 8 \int_0^1 (\sqrt{2z} - z^2) dz \left(\int_0^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} d(\sin \theta) \right)$$

$$+ 8 \sqrt{2} \int_1^2 z dz \int_{\sqrt{z-1}}^1 \sqrt{1 - y^2} dy - 8 \int_1^2 z^2 dz \int_{\sqrt{1-\frac{1}{2}}}^1 \sqrt{1 - y^2} dy$$

$$= 2\pi \left(\frac{\sqrt{2}}{2} - \frac{1}{3} \right) + 2\sqrt{2} \int_1^2 z \left(0 - \sqrt{z-1} \sqrt{2-z} + 2 \times \left(\frac{\pi}{2} - \arccos \sqrt{z-1} \right) \right) dz - 2 \int_1^2 z^2 \left(0 - \sqrt{\left(2 - \frac{z}{2}\right) \left(\frac{z}{2} - 1\right)} + 2 \times \left(\frac{\pi}{2} - \arccos \sqrt{\frac{z}{2}} \right) \right) dz$$

$$= 2\pi \left(\frac{\sqrt{2}}{2} - \frac{1}{3} \right) + 4\sqrt{2} \int_1^2 z \arccos \sqrt{z-1} dz - 4 \int_1^2 z^2 \arccos \sqrt{2 - \frac{z}{2}} dz$$

$$\text{設 } y = \arccos \sqrt{z-1}, \quad z \in [1, 2] \Leftrightarrow z-1 = \cos^2 y, \quad y \in [0, \frac{\pi}{2}]$$

$$\int_1^2 z \arccos \sqrt{z-1} dz = \int_{\frac{\pi}{2}}^0 (1 + \cos^2 y) y (-\sin 2y) dy = \int_0^{\frac{\pi}{2}} \frac{3 + \cos 2y}{2} y \cdot \sin 2y dy$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (3y \sin 2y + \frac{y}{2} \sin 4y) dy = \frac{3}{4} \int_0^{\frac{\pi}{2}} y d(-\cos 2y) + \frac{1}{16} \int_0^{\frac{\pi}{2}} y d(-\cos 4y)$$

$$= -\frac{3}{4} \left(y \cos 2y \right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \cos 2y dy - \frac{1}{16} \left(y \cos 4y \right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \cos 4y dy = \frac{3\pi}{8} - \frac{\pi}{32} + \frac{3}{8} \sin 2y \Big|_0^{\frac{\pi}{2}} + \frac{1}{64} \sin 4y \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{11\pi}{32}$$

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$$\text{所以 } \int dx dy dz = \sqrt{2}\pi - \frac{2}{3}\pi + \frac{11\sqrt{2}\pi}{8} - 4 \int_1^2 z^2 \arccos \sqrt{2-\frac{z}{2}} dz$$

$$\text{設 } x = \arccos \sqrt{2-\frac{z}{2}}, z \in [1, 2] \Leftrightarrow 2-\frac{z}{2} = \cos^2 x, x \in [0, \frac{\pi}{2}] \Leftrightarrow z = \frac{2}{2-\cos^2 x}, x \in [0, \frac{\pi}{2}]$$

$$dz = -\frac{2}{(2-\cos^2 x)^2} \sin 2x dx$$

$$\int_1^2 z^2 \arccos \sqrt{2-\frac{z}{2}} dz = \int_{\frac{\pi}{2}}^0 \left(\frac{2}{2-\cos^2 x}\right)^2 x \cdot \left(-\frac{2}{(2-\cos^2 x)^2}\right) \sin 2x dx$$

$$= 8 \int_0^{\frac{\pi}{2}} \frac{1}{(2-\cos^2 x)^4} x \sin 2x dx = 4 \int_0^{\frac{\pi}{2}} \frac{x}{\left(\frac{3-\cos 2x}{2}\right)^4} d(-\cos 2x)$$

$$= 64 \int_0^{\frac{\pi}{2}} x d\left(-\frac{1}{3} \cdot \frac{1}{(3-\cos 2x)^3}\right) = -\frac{64}{3} \int_0^{\frac{\pi}{2}} x d\left(\frac{1}{(3-\cos 2x)^3}\right) = -\frac{64}{3} \left(\frac{x}{(3-\cos 2x)^3}\right) \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \frac{1}{(3-\cos 2x)^3} dx$$

$$= -\frac{\pi}{6} + \frac{64}{3} \int_0^{\frac{\pi}{2}} \frac{1}{(3-\cos 2x)^3} dx$$

$$\text{設 } y = \tan x, dx = \frac{1}{y^2+1} dy, \int_0^{\frac{\pi}{2}} \frac{1}{(3-\cos 2x)^3} dx = \int_0^{\frac{\pi}{2}} \left(\frac{1+\tan^2 x}{2+4\tan^2 x}\right)^3 dx = \int_0^{+\infty} \left(\frac{1+y^2}{2+4y^2}\right)^3 \cdot \frac{1}{1+y^2} dy$$

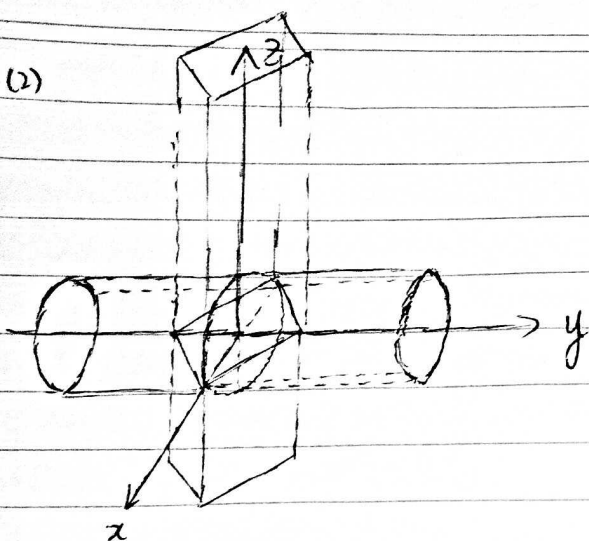
$$= \int_0^{+\infty} \frac{(1+y^2)^2}{8(1+2y^2)^3} dy = \frac{1}{64} \int_0^{+\infty} \frac{(y^2+1)^2}{(y^2+\frac{1}{2})^3} dy = \frac{1}{64} \int_0^{+\infty} \left(\frac{1}{y^2+\frac{1}{2}} + \frac{1}{(y^2+\frac{1}{2})^2} + \frac{1}{4(y^2+\frac{1}{2})^3}\right) dy$$

$$= \frac{1}{64} \int_0^{+\infty} \left(\frac{2}{(\sqrt{2}y)^2+1} + \frac{4}{[(\sqrt{2}y)^2+1]^2} + \frac{2}{((\sqrt{2}y)^2+1)^3}\right) dy$$

$$= \frac{\sqrt{2}}{64} \times \left(2 \arctan(\sqrt{2}y) \Big|_0^{+\infty} + 2 \times \frac{\pi}{4}\right) = \frac{\sqrt{2}}{64} \cdot \left(\pi + \frac{\pi}{2}\right) = \frac{3\sqrt{2}}{128} \pi$$

$$\text{所以 } \int dx dy dz = \frac{19\sqrt{2}}{8}\pi - \frac{2}{3}\pi - 4 \cdot \left(-\frac{\pi}{6} + \frac{\sqrt{2}}{2}\pi\right) = \frac{3\sqrt{2}}{8}\pi$$

(2)



V 在 xy 平面上的投影是 $D = \{(x, y, 0) \mid |x| + |y| \leq a\}$.
 $-\sqrt{a^2 - x^2} \leq z \leq \sqrt{a^2 - x^2}$

$$\int dx dy dz = 2 \int_D \sqrt{a^2 - x^2} dx dy = 2 \int_{-a}^a dx \int_{|x|-a}^{a-|x|} \sqrt{a^2 - x^2} dy$$

$$= 2 \int_{-a}^a (a - |x|) \sqrt{a^2 - x^2} dx$$

$$= 4 \int_0^a (a - x) \sqrt{a^2 - x^2} dx$$

$$= 4a^3 \int_0^a \left(1 - \frac{x}{a}\right) \sqrt{1 - \left(\frac{x}{a}\right)^2} d\left(\frac{x}{a}\right)$$

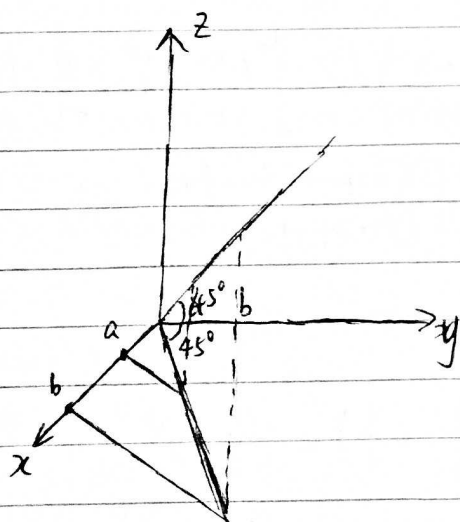
$$= 4a^3 \int_0^{\frac{\pi}{2}} (1 - \sin \theta) \cos \theta d \sin \theta$$

$$= 4a^3 \int_0^{\frac{\pi}{2}} (\cos^2 \theta - \sin \theta \cos^2 \theta) d\theta$$

$$= 4a^3 \cdot \left(\frac{\pi}{4} + \frac{1}{3} \cos^3 \theta \right) \Big|_0^{\frac{\pi}{2}} = 4a^3 \cdot \left(\frac{\pi}{4} - \frac{1}{3} \right)$$

3. 證: 設 $D = \{(x, y, z) \mid a \leq x \leq b, a \leq y \leq x, a \leq z \leq y\} = \{(x, y, z) \mid a \leq z \leq y \leq x \leq b\}$.
 $= \{(x, y, z) \mid a \leq z \leq b, z \leq y \leq b, y \leq x \leq b\}$.

$$\text{所以 } \int_a^b dx \int_a^x dy \int_a^y f(x, y, z) dz = \int_D f(x, y, z) d\mu = \int_a^b dz \int_z^b dy \int_y^b f(x, y, z) dx$$



$$4. \int_0^a dx \int_0^x dy \int_0^y f(x) f(y) f(z) dz = \int_0^a dz \int_z^a dy \int_y^a f(x) f(y) f(z) dx = \int_0^a dy \int_0^y dz \int_y^a f(x) f(y) f(z) dx$$

$$= \int_0^a dx \int_0^x dz \int_z^x f(x) f(y) f(z) dy = \int_0^a dz \int_z^a dx \int_z^x f(x) f(y) f(z) dy = \int_0^a dy \int_y^a dx \int_0^y f(x) f(y) f(z) dz$$

培道中學校

姓名：

(號)

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因此，我們把 x, y, z 進行輪換變元，有

$$A = \int_0^a dx \int_0^x dy \int_0^y f(x) f(y) f(z) dz = \int_0^a dx \int_x^a dy \int_y^a f(x) f(y) f(z) dz = \int_0^a dx \int_0^x dy \int_x^a f(x) f(y) f(z) dz$$

$$= \int_0^a dx \int_0^x dy \int_y^a f(x) f(y) f(z) dz = \int_0^a dx \int_x^a dy \int_x^y f(x) f(y) f(z) dz = \int_0^a dx \int_x^a dy \int_0^x f(x) f(y) f(z) dz$$

$$2A = \int_0^a dx \int_0^x dy \int_0^x f(x) f(y) f(z) dz = \int_0^a dx \int_x^a dy \int_x^a f(x) f(y) f(z) dz$$

$$3A = \int_0^a dx \int_0^x dy \int_0^a f(x) f(y) f(z) dz = \int_0^a dx \int_x^a dy \int_0^a f(x) f(y) f(z) dz$$

$$6A = \int_0^a dx \int_0^a dy \int_0^a f(x) f(y) f(z) dz = \left(\int_0^a f(x) dx \right) \left(\int_0^a f(y) dy \right) \left(\int_0^a f(z) dz \right) = \left(\int_0^a f(x) dx \right)^3$$

$$\Rightarrow A = \frac{1}{6} \left(\int_0^a f(t) dt \right)^3$$

$$\begin{aligned} 5. \int_0^a dx \int_0^x dy \int_0^y f(z) dz &= \int_0^a dz \int_z^a dy \int_y^a f(z) dx = \int_0^a dz \int_z^a f(z) (a-y) dy = \int_0^a f(z) \left(ay - \frac{1}{2} y^2 \right) \Big|_z^a dz \\ &= \int_0^a f(z) \left(a^2 - \frac{1}{2} a^2 - az + \frac{1}{2} z^2 \right) dz = \frac{1}{2} \int_0^a (a-z)^2 f(z) dz = \frac{1}{2} \int_0^a (a-t)^2 f(t) dt \end{aligned}$$

6. 由積分中值定理知， $\forall r > 0, \exists P(\xi_x, \xi_y, \xi_z) \in B_r(0,0,0)$

$$\text{使得 } \underbrace{\iiint_{(x,y,z) \in B_r(0,0,0)} f(x,y,z) dx dy dz}_{(x,y,z) \in B_r(0,0,0)} = f(\xi_x, \xi_y, \xi_z) \underbrace{\iiint_{(x,y,z) \in B_r(0,0,0)} dx dy dz}_{(x,y,z) \in B_r(0,0,0)} = \frac{4}{3} \pi r^3 f(\xi_x, \xi_y, \xi_z)$$

$$\Rightarrow \frac{3}{4\pi r} \iiint_{x^2+y^2+z^2 \leq r^2} f(x,y,z) dx dy dz = f(\xi_x, \xi_y, \xi_z)$$

令 $r \rightarrow 0^+$ ，則 $(\xi_x, \xi_y, \xi_z) \rightarrow (0,0,0)$ ，由 f 的連續性知，所求極限是 $f(0,0,0)$

$$7. (1) \iiint_{x^2+y^2+z^2 \leq 2} \sqrt{x^2+y^2+z^2} dx dy dz = \int_0^{\sqrt{2}} dr \int_0^{\pi} d\theta \int_0^{2\pi} |r^2 \sin \theta| d\varphi = 2\pi \left(\int_0^{\sqrt{2}} r^3 dr \right) \int_0^{\pi} \sin \theta d\theta$$

$$= 2\pi \cdot \frac{(\sqrt{2})^4}{4} \cdot 2 = 4\pi.$$

$$(2) \iiint_D (x^2+y^2) dx dy dz = \int_0^{2\pi} d\varphi \int_0^{\frac{\pi}{2}} d\theta \int_a^b r^2 \sin \theta |r^2 \sin \theta| dr = 2\pi \left(\int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta \right) \left(\int_a^b r^4 dr \right)$$

$$= 2\pi \left(\int_0^{\frac{\pi}{2}} (1 - \cos^2 \theta) d\theta \right) \cdot \frac{1}{5} (b^5 - a^5) = 2\pi \left(-\cos \theta \Big|_0^{\frac{\pi}{2}} + \frac{1}{3} \cos^3 \theta \Big|_0^{\frac{\pi}{2}} \right) \cdot \frac{1}{5} (b^5 - a^5)$$

$$= \frac{4\pi}{15} (b^5 - a^5)$$

$$(3) \text{取 } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ar \sin \theta \cos \varphi \\ br \sin \theta \sin \varphi \\ cr \cos \theta \end{pmatrix}, \quad \frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = abc r^2 \sin \theta$$

$$\iiint_D \left(1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \right)^{\frac{1}{2}} dx dy dz = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \int_0^1 (1-r^2)^{\frac{1}{2}} \cdot abc |r^2 \sin \theta| dr$$

$$= 2\pi abc \left(\int_0^{\pi} \sin \theta d\theta \right) \left(\int_0^1 \sqrt{1-r^2} r^2 dr \right) = 4\pi abc \left(\int_0^{\frac{\pi}{2}} \cos^2 \theta \sin^2 \theta d\theta \right)$$

$$= \pi abc \int_0^{\frac{\pi}{2}} \sin^2 2\theta d\theta = \frac{\pi}{2} abc \int_0^{\frac{\pi}{2}} (1 - \cos 4\theta) d\theta = \frac{\pi^2}{4} abc$$

$$8. (1) \text{令 } \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}, \quad \text{其中 } |u| \leq h_1, |v| \leq h_2, |w| \leq h_3$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}^{-1} \begin{pmatrix} u \\ v \\ w \end{pmatrix}, \quad \frac{\partial(x,y,z)}{\partial(u,v,w)} = \left| \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \right|^{-1}$$

$$\int dx dy dz = \iiint_{(u,v,w) \in [-h_1, h_1] \times [-h_2, h_2] \times [-h_3, h_3]} \left| \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \right|^{-1} du dv dw = 8h_1 h_2 h_3 \left| \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \right|^{-1}$$

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(3) 設 $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \sin \theta \cos \varphi \\ r \sin \theta \sin \varphi \\ r \cos \theta \end{pmatrix}$, $r^4 \leq a^3 r \cos \theta \Leftrightarrow r^3 \leq a^3 \cos \theta \Leftrightarrow 0 \leq r \leq a \sqrt[3]{\cos \theta}$

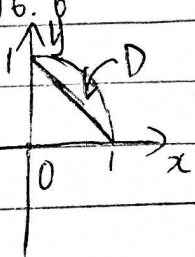
$$\begin{aligned} \int_V dx dy dz &= \int_0^{2\pi} d\varphi \int_0^{\frac{\pi}{2}} d\theta \int_0^{a \sqrt[3]{\cos \theta}} r^2 \sin \theta dr = 2\pi \int_0^{\frac{\pi}{2}} \frac{1}{3} (a \sqrt[3]{\cos \theta})^3 \sin \theta d\theta = \frac{a^3}{3} \pi \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta \\ &= \frac{a^3}{3} \pi \cos 2\theta \Big|_{\frac{\pi}{2}}^0 = \frac{2a^3}{3} \pi \end{aligned}$$

(5) 設 $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a r \sin \theta \cos \varphi \\ b r \sin \theta \sin \varphi \\ c r \cos \theta \end{pmatrix}$, $r^4 \leq r^2 \sin^2 \theta \Leftrightarrow r^2 \leq \sin^2 \theta \Leftrightarrow 0 \leq r \leq |\sin \theta| = \sin \theta$

$$\begin{aligned} \int_V dx dy dz &= \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \int_0^{\sin \theta} abc r^2 \sin \theta dr = 2\pi abc \int_0^{\pi} \frac{1}{3} \sin^4 \theta d\theta \\ &= \frac{2\pi}{3} abc \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right)^2 d\theta = \frac{\pi}{6} abc \int_0^{\pi} (1 - 2\cos 2\theta + \frac{1 + \cos 4\theta}{2}) d\theta \\ &= \frac{\pi}{6} abc \cdot \left(\frac{3\pi}{2} - \sin 2\theta \Big|_0^{\pi} + \frac{1}{4 \times 2} \sin 4\theta \Big|_0^{\pi} \right) = \frac{\pi^2}{4} abc \end{aligned}$$

問題 16.6

1.



$$\begin{aligned} \iint_D \frac{dx dy}{xy(\log^2 x + \log^2 y)} &= \int_0^1 dx \int_{1-x}^{\sqrt{1-x^2}} \frac{dy}{xy(\log^2 x + \log^2 y)} \\ &= \int_0^1 dx \int_{1-x}^{\sqrt{1-x^2}} \frac{1}{x \log x} \cdot \frac{1}{1 + \left(\frac{\log y}{\log x}\right)^2} d\left(\frac{\log y}{\log x}\right) = \int_0^1 \frac{1}{x \log x} \cdot \arctan\left(\frac{\log y}{\log x}\right) \Big|_{1-x}^{\sqrt{1-x^2}} dx \\ &= \int_0^1 \frac{1}{\log x} \left(\arctan\left(\frac{\frac{1}{2} \log(1-x^2)}{\log x}\right) - \arctan\left(\frac{\log(1-x)}{\log x}\right) \right) d \log x \end{aligned}$$